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CHE 457

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Problem Statement for the class

One strong example problem from the paper that could be taught in a graduate transport phenomena class is the derivation and application of the Marangoni spreading scaling laws for an ink/alcohol droplet spreading on a liquid substrate. Specifically, the problem involving the two experimentally observed spreading regimes with droplet radius being used for scaling as $R = t^{3/8}$ and $R = t^{1/4}$ to make an example that combines interfacial transport, viscous flow, scaling analysis, and dimensional reasoning.

The problem could be presented by asking students to derive the spreading laws from a balance between Marangoni stresses and viscous stresses. In the paper, the authors show that when the viscous penetration depth is smaller than the substrate thickness, the characteristic spreading radius follows where A is the surface activity, M is, the surfactant mass, ρ is density, and η is viscosity.

$$R = \left(\frac{A^2 M^2}{\rho \eta} \right)^{\frac{1}{8}} \times t^{\frac{3}{8}}$$

When the substrate thickness dominates the vertical length scale, the spreading instead follows with H representing the substrate thickness.

$$R = \left(\frac{A M H}{\eta} \right)^{\frac{1}{4}} \times t^{\frac{1}{4}}$$

This example is pedagogically valuable because it demonstrates how transport phenomena principles can be used to predict experimentally measurable behavior without solving the full Navier–Stokes equations. Students would practice identifying dominant physical mechanisms, selecting appropriate characteristic scales, and applying scaling arguments to complex multiphase flow systems. The problem also introduces important graduate-level concepts such as Marangoni convection, thin-film hydrodynamics, interfacial instability, and rheological effects in non-Newtonian fluids. Additionally, the connection between the theoretical scaling laws and the experimental data collapse shown in the paper provides an excellent example of how transport theory is validated experimentally.

Consider an insoluble surfactant droplet of mass M spreading on a liquid substrate due to surface-tension gradients or Marangoni force spreading the goal is to derive the two spreading laws observed experimentally in the paper.

Problem:

Derive the scaling laws for the spreading radius $R(t)$ of a surfactant-laden droplet on a liquid substrate by balancing:

- Marangoni stress
- Viscous stress

Then solve for:

1. For a deep liquid layer:

$$R = \left(\frac{A^2 M^2}{\rho \eta} \right)^{1/8} t^{3/8}$$

2. For a thin liquid layer of thickness H :

$$R = \left(\frac{AMH}{\eta} \right)^{1/4} t^{1/4}$$

Solution:

To solve for the two cases we must first set up the equation and basis for the derivation:

By setting area as the radius of the function of spreading, we can apply the concentration spread of the drop,

$$Area = R^2$$

$$R^2 \times \Gamma = M$$

Then we rewrite this in to:

$$\Gamma = \frac{M}{R^2}$$

Then we define the equation for the Marangoni stress:

To imply Marangoni stress, we must use the surface tension gradient to shear stress:

$$\tau_M = \frac{AF}{R}$$

When we apply the spreading to the gradient:

$$\Gamma = \frac{M}{R^3}$$

This turns onto:

$$\tau_M = \frac{AM}{R^3}$$

Then we must determine the viscous stress:

$$\tau_v = \eta \gamma$$

Using scaling for shear rate by having U as the radial velocity and δ_c as the depth.

$$\gamma = \frac{U}{\delta_c}$$

This will give us the following equation:

$$\tau_v = \eta \frac{U}{\delta_c}$$

Now we must estimate the velocity of the scale through the spread of droplet as R as a function of time t :

$$U = \frac{R}{t}$$

Now we add it to the viscous stress:

$$\tau_v = \eta \frac{\frac{R}{t}}{\delta_c}$$

This gives us the stress resisting the spread:

$$\tau_v = \eta \frac{R}{t \times \delta_c}$$

Now to finally derive the equation we must balance out all the stress equations into one uniform equation:

we start with:

$$\tau_M = \tau_v$$

This is expanded into the following:

$$\frac{AM}{R^3} = \eta \frac{R}{t \times \delta_c}$$

Now we multiply R on both sides of the equation:

$$AM = \eta \frac{R^4}{t \times \delta_c}$$

We rearrange this into:

$$R^4 = \frac{AM \times \delta_c}{\eta} t$$

Now we that the fourth route to get the spreading equation of the paper:

$$R = \left(\frac{AM \times \delta_c}{\eta} t \right)^{\frac{1}{4}}$$

Now we move to do the first case and derive the formula to the first liquid layer.

We first use the depth function and tie it to the viscous diffusion:

$$\delta_c = \sqrt{\frac{\psi}{\rho}}$$

Now we apply this to the general equation from before:

$$R = \left(\frac{AM \times \sqrt{\frac{\psi}{\rho}}}{\eta} t \right)^{\frac{1}{4}}$$

Now we combine the time (t) terms to make the function more simplified:

$$t \times \sqrt{t} = t^{\frac{3}{2}}$$

Then we add it into the general equation:

$$R = \left(\frac{AM}{\eta} \times \sqrt{\eta} \times \frac{t^{\frac{3}{2}}}{\sqrt{\rho}} \right)^{\frac{1}{4}}$$

Now we simplify the viscosity terms within the equation:

$$R = (AM\eta^{-\frac{1}{2}}\rho^{-\frac{1}{2}}t^{\frac{3}{2}})^{\frac{1}{4}}$$

Now we apply simplify by applying $\frac{1}{4}$ as a power law:

$$R = A^{\frac{1}{4}} \times M^{\frac{1}{4}} \times \eta^{-\frac{1}{8}} \times \rho^{-\frac{1}{8}} \times t^{\frac{3}{8}}$$

Now we simplify and rearrange as an equation of R to give us the deep layer Marangoni spreading force law:

$$R = \left(\frac{A^2 M^2}{\rho \eta} \right)^{\frac{1}{2}} t^{\frac{3}{8}}$$

Now we move on to case 2 where the equation must be formulated into the application of thin film spreading.

To do this we must use the general spread equation that was derived:

$$R = (\frac{AM \times \delta_c}{\eta} t)^{\frac{1}{4}}$$

And convert the depth δ_c into a formulation of height H:

$$\delta_c = H$$

This turns into the equation needed for the thin fil solving the second case:

$$R = (\frac{AM \times H}{\eta} t)^{\frac{1}{4}}$$

We now expand this to get the true final answer:

$$R = (\frac{AMH}{\eta})^{\frac{1}{4}} t^{\frac{1}{4}}$$

References:

- Chan, S. T., & Fried, E. (2024). *Marangoni spreading on liquid substrates in new media art. PNAS Nexus*, 3(2), 1–12. <https://doi.org/10.1093/pnasnexus/pgae059> (Chan & Fried, 2024)